1 Basics

Taylor: $f(a)+\frac{f'(a)}{1!}(x-a)+\frac{f''(a)}{2!}(x-a)^2+\dots$
$f(a)+\left.\frac{\partial f(x)}{\partial x}\right|_a-\frac{1}{2}(x-a)^\top\left.\frac{\partial^2 f(x)}{\partial x\partial x^\top}\right|_a(x-a)$
Power series of exp: $\exp(x)\triangleq\sum_{k=0}^{\infty}\frac{x^k}{k!}$
Entropy: $H(X)=\mathbb{E}[X]-\log P(X=x)$
Divergence: $\mathrm{KL}(p\|q)=\sum_{x\in\mathcal{X}}p(x)\log\frac{p(x)}{q(x)}\geq 0$
CE: $\mathrm{CE}(p\|q):=-\sum_{x\in\mathcal{X}}p(x)\log q(x)=\mathrm{KL}(p\|q)+H(p)$
Cauchy-Schwarz: $|\mathbb{E}X,Y|^2\leq\mathbb{E}X^2\mathbb{E}Y^2$
Jensen, $f(X)$ convex: $f(\mathbb{E}X)\leq\mathbb{E}f(X)$

1.1 Probability / Statistics

Gaussian: $\mathcal{N}(x|\mu,\sigma^2)=\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{1}{2}\frac{(x-\mu)^2}{\sigma^2}\right)$
$\mathcal{N}(x|\mu,\Sigma)=\frac{1}{\sqrt{(2\pi)^d|\Sigma|}}\exp\left(-\frac{1}{2}(x-\mu)^\top\Sigma^{-1}(x-\mu)\right)$
$X\sim\mathcal{N}(\mu,\Sigma),Y=A+BX\Rightarrow Y\sim\mathcal{N}(A+B\mu,B\Sigma B^\top)$
Binomial: $f(k,n;p)=P(X=k)=\binom{n}{k}p^k(1-p)^{n-k}$
$\mathbb{V}X=\mathbb{E}(X-\mathbb{E}X)^2=\mathbb{E}X^2-\mathbb{E}X^2$
$\mathbb{V}X+Y=\mathbb{V}X+\mathbb{V}Y+2\mathrm{Cov}(X,Y)$
$\mathrm{Cov}(X,Y)=\mathbb{E}(X-\mathbb{E}X)(Y-\mathbb{E}Y)=\mathbb{E}XY-\mathbb{E}X\mathbb{E}Y$
$\mathrm{Cov}(aX,bY)=ab\mathrm{Cov}(X,Y)$

1.2 Calculus

$\int uv'dx=uv-\int u'vdx$
$\frac{\partial}{\partial x}\frac{g}{h}=\frac{g'h}{h^2}-\frac{gh'}{h^2}$
$\frac{\partial}{\partial x}(b^\top Ax)=A^\top b$
$\frac{\partial}{\partial x}(b^\top x)=\frac{\partial}{\partial x}(x^\top b)=b$
$\frac{\partial}{\partial X}(c^\top X^\top b)=bc^\top$
$\frac{\partial}{\partial X}(c^\top Xb)=cb^\top$
$\frac{\partial}{\partial x}(x^\top Ax)=(A^\top+A)x\xrightarrow{A\text{ sym.}}2Ax$
$\frac{\partial}{\partial X}\mathrm{Tr}(X^\top A)=A$
Trace trick: $x^\top Ax=\mathrm{Tr}(x^\top Ax)=\mathrm{Tr}(xx^\top A)=\mathrm{Tr}(Axx^\top)$
$|X^{-1}|=|X|^{-1}$
$\frac{\partial}{\partial X}\log|X|=X^{-\top}$
$\frac{d}{dx}|x|=\frac{x}{|x|}$
$\frac{\partial}{\partial x}\|x\|^2=\frac{\partial}{\partial x}(x^\top x)=2x$
$\frac{\partial}{\partial x}\|x-b\|^2=\frac{x-b}{\|x-b\|^2}$
$\frac{\partial}{\partial x}\|x\|_1=\mathrm{sgn}(x)$, $\mathrm{sgn}(x)\in\{\pm1\}^p$ is row-wise
$\sigma(x)=\frac{1}{1+\exp(-x)}\Rightarrow\nabla\sigma(x)=\sigma(x)(1-\sigma(x))$
$\tanh x=\frac{2\sinh x}{2\cosh x}=\frac{e^x-e^{-x}}{e^x+e^{-x}}$
$\nabla\tanh x=1-\tanh^2 x$

2 Density Estimation

Bayesianism: Define prior $P(\theta)$, define likelihood $P(X|\theta)$, compute posterior $P(\theta|x_1\dots n)$.
Bayes: $P(\theta|X)=\frac{P(X|\theta)P(\theta)}{P(X)}$, $P(X)=\sum_\theta P(X|\theta_i)P(\theta_i)$
Frequentism: Define param. model $P(Y|X,\theta)$, compute likelihood of data $P(X,Y|\theta)$ and compute $\hat{\theta}_{\mathrm{MLE}}$ via argmax_θ of likelihood.
Ex. Gaussian: $X_{n_i=1}\sim\mathcal{N}(\mu,\sigma^2)\Rightarrow\log L_n(\mu,\sigma)=$
$-n\log\sigma-\frac{nS^2}{2\sigma^2}-\frac{n(\bar{X}-\mu)^2}{2\sigma^2}\Rightarrow\hat{\mu}=\bar{X},\hat{\sigma}=S$

2.1 Estimation - MLE Properties

Consistency: $\forall\epsilon>0,P\{|\hat{\theta}_n-\theta^*|>\epsilon\}\xrightarrow{n\to\infty}0$
Equivariance: If $\hat{\theta}_n$ is MLE of θ, then $g(\hat{\theta}_n)$ is MLE of $g(\theta)$.
Asymp. normality: $\frac{1}{\sqrt{n}}(\hat{\theta}_n-\theta^*)\to\mathcal{N}(0,J^{-1}(\theta^*)I_n(\theta^*)J^{-1}(\theta^*))$
Asymp. efficiency: $\hat{\theta}_n$ minimises $\mathbb{E}(\hat{\theta}_n-\theta^*)^2$ as $n\to\infty$, i.e. $\mathbb{E}(\hat{\theta}_n-\theta^*)^2\xrightarrow{n\to\infty}\frac{1}{I_n(\theta^*)}$ (Rao Cr.)
Among all consistent estimators $\hat{\theta}_n$ has smallest variance: $\lim_{n\to\infty}(\mathbb{V}\hat{\theta}_nI_n(\theta^*))^{-1}=1$

2.2 Rao Cramer inequality

All $\mathbb{E}$ w.r.t. $P(x|\theta^*)$
Score func.: $\Lambda=\frac{\partial\log P(x|\theta)}{\partial\theta}$, $\mathbb{E}\Lambda=0$
Fisher info.: $I_n(\theta)=\mathbb{V}\Lambda=\mathbb{E}\Lambda^2$
$\mathbb{E}[x|\theta]\Lambda\hat{\theta}=\frac{\partial b_{\hat{\theta}}}{\partial\theta}+1$ (1) where $b_{\hat{\theta}}:=\mathbb{E}[x|\theta]\hat{\theta}-\theta$.
$\mathbb{E}[x|\theta](\Lambda-\mathbb{E}\Lambda)(\hat{\theta}-\mathbb{E}\hat{\theta})^2=(\mathbb{E}[x|\theta]\Lambda\hat{\theta}-\mathbb{E}[x|\theta]\Lambda\mathbb{E}\hat{\theta})^2=(\mathbb{E}[x|\theta]\Lambda\hat{\theta})^2$ (2)
Cauchy-Schwarz: $\mathbb{E}[x|\theta]\Lambda\hat{\theta}^2=(\mathbb{E}\Lambda(\hat{\theta}-\mathbb{E}\hat{\theta}))^2\leq$
$\mathbb{E}[x|\theta]\Lambda^2\cdot\mathbb{E}[x|\theta](\hat{\theta}-\theta-\mathbb{E}\hat{\theta}+\theta)^2=\mathbb{E}[x|\theta]\Lambda^2(\mathbb{E}[x|\theta](\hat{\theta}-\theta)^2-b_{\hat{\theta}}^2)$
General bound: $\mathbb{E}(\hat{\theta}_n-\theta^*)^2\geq\frac{(1+\frac{\partial}{\partial\theta}b_{\hat{\theta}})^2}{\mathbb{E}\Lambda^2}+b_{\hat{\theta}}^2$
Unbiased case: $\mathbb{E}(\hat{\theta}_n-\theta^*)^2=\mathbb{V}\hat{\theta}_n\geq\frac{1}{I_n(\theta^*)}$
Tradeoff: $\mathbb{E}(\hat{\theta}_n-\theta^*)^2=\mathbb{V}\hat{\theta}_n+\mathrm{bias}^2(\hat{\theta}_n)$

3 (Linear) Regression

Model: $\hat{y}=X\beta$
Bayesian view: $(Y|X,\beta)\sim\mathcal{N}(x^\top\beta,\sigma^2I)$
Distrib. of estimator $\hat{\beta}_{\mathrm{LS}}\sim\mathcal{N}(\beta,(X^\top X)^{-1}\sigma^2)$
Ridge: $\mathrm{RSS}(\beta,\lambda)=(y-X^\top\beta)^\top(y-X^\top\beta)+\lambda\beta^\top\beta$
$\hat{\beta}=(X^\top X+\lambda I)^{-1}X^\top y$, prior: $\beta\sim\mathcal{N}(0,\frac{\sigma^2}{\lambda}I)$
Lasso: $\hat{\beta}=\mathrm{argmin}_\beta\sum_{i\leq n}(y_i-x_i^\top\beta)^2+\lambda\|\beta\|_1$
(no closed form), prior: $p(\beta_i)=\frac{\lambda}{4\sigma^2}\exp(-|\beta_i|\frac{\lambda}{2\sigma^2})$
Bias-variance: $\mathrm{MSE}=\mathbb{E}[D,X,Y](\hat{f}(x)-Y)^2$
$=(\mathbb{E}[D,X]\hat{f}(x)-\mathbb{E}Y)^2+\mathbb{E}[D](\mathbb{E}[D,X]\hat{f}(x)-\hat{f}(x))^2+\mathbb{E}(E[Y]-Y)^2$
$=\mathrm{bias}^2+\mathrm{variance}+\mathrm{noise}$
Non-Linear: $\hat{\beta}=(\Phi^\top\Phi)^{-1}\Phi^\top Y$, $y'=\phi(x')^\top\hat{\beta}$
Gauss-Markov Theorem: For any linear estimator $\tilde{\theta}=c^\top y=a^\top(\hat{\beta}+Dy)$ that is unbiased for $a^\top\beta$, it holds: $\mathbb{V}a^\top\hat{\beta}\leq\mathbb{V}c^\top y$.

3.1 Causality

Shortcut Learning: Model uses spurious correlation $X\leftrightarrow Y$ (e.g. grass $\leftrightarrow$ cow)
$X^{\perp\alpha}$: Part of X not causally influenced by α
Causal: $X^{\perp W}$ and $X,W\&Y$ are causes of Y.
Anti-Causal: $X^{\perp W}$ and $X,W\&Y$ are caused by Y.
W non-causal feature (e.g. background) influences X but not Y.
Counterfactually Inv.: $f(X(w))=f(X(w'))$ for $w,w'\in W$.
Confounding: Hidden U affects W,X.
Selection: Hidden S filters on W,X.
Theorem For Counterfactual Invariance (CI):
Anti-causal: $f(X)\perp W|Y$
Causal (no selection): $f(X)\perp W$
Causal (no confounding): $Y\perp X|X^{\perp W},W$ and $f(X)\perp W|Y$.

4 Uncertainty Quantification

4.1 Bayesian Linear Regression

Model: $y=X^\top\beta+\epsilon$, with $\epsilon\sim\mathcal{N}(\epsilon|0,\sigma^2I)$
Likelihood: $P(y|X,\beta,\sigma)=\mathcal{N}(y|X^\top\beta,\sigma^2I)$
Prior: $P(\beta|\Lambda)=\mathcal{N}_d(\beta|0,\Lambda^{-1})\propto\exp(-\frac{1}{2}\beta^\top\Lambda\beta)$
(Ridge regr. if $\Lambda=\lambda I$ and $\sigma=1$)
Posterior: $P(\beta|X,y,\Lambda)=\mathcal{N}(\beta|\mu_\beta,\Sigma_\beta)$
with $\mu_\beta=(X^\top X+\sigma^2\Lambda)^{-1}X^\top y$
and $\Sigma_\beta=\sigma^2(X^\top X+\sigma^2\Lambda)^{-1}$
$\Rightarrow y\sim\mathcal{N}(y|0,X\Lambda^{-1}X^\top+\sigma^2I)$, $k(x_i,x_j)\triangleq x_i^\top\Lambda^{-1}x_j$

4.2 Gaussian Process $O(n^3)$

$y\sim\mathcal{N}(y|m(X),K(X,X)+\sigma^2I)$
$\begin{bmatrix}y\\y_{n+1}\end{bmatrix}\sim\mathcal{N}\left(\begin{bmatrix}y\\y_{n+1}\end{bmatrix}\middle|\begin{bmatrix}m(X)\\m(x_{n+1})\end{bmatrix},\begin{bmatrix}C_n&k\\k^\top&c\end{bmatrix}\right)$
$p(y_{n+1}|x_{n+1},X,y)=\mathcal{N}(y_{n+1}|\mu_{n+1},\sigma_{n+1}^2)$
with $\mu_{n+1}=m(x_{n+1})+k^\top C_n^{-1}(y-m(X))$
and $\sigma_{n+1}^2=c-k^\top C_n^{-1}k$
where $K=k(X,X)$, $k=k(x_{n+1},X)$,
$C_n=K+\sigma^2I$, $c=k(x_{n+1},x_{n+1})+\sigma^2$

4.3 Kernels

Scalar product $K_{ij}=k(x_i,x_j)$
Valid kernel: must be symmetric and p.s.d. ($x^\top Kx\geq0\,\forall x$ or pos. eigenvalues or pos. principal minors). Must have a (pot. ∞-dim.) feature vector ϕ s.t. $k(x,x')=\phi(x)^\top\phi(x')$.
Common kernels:
Polynomial: $(x^\top x'+1)^p$, $p\in\mathbb{N}$
RBF (Gaussian): $\sigma^2\exp(-\|x-x'\|_2^2/2l^2)$
Periodic: $\sigma^2\exp(-2\sin^2(\pi\|x-x'\|/p)/l^2)$
Kernel construction:

- k_1+k_2
- $c\cdot k_1$, $c>0$
- $k_1\cdot k_2$
- $f(x)k_1(x,x')f(x')$
- $k(\phi(x),\phi(x'))$ with $\phi:\mathcal{X}\to\mathbb{R}^d$
- $g(k_1)$ with g : exp. or polyn. w/ all pos. coeff.

4.4 Statistical Model Validation

Bootstrap: $\hat{R}_{\mathrm{bs}}=\frac{1}{n}\sum_{i=1}^n\frac{1}{|C_{-i}|}\sum_{b\in C_{-i}}L(y_i,\hat{f}_{*b}(x_i))$
where $C_{-i}=\{b:(x_i,y_i)\notin Z_{*b}\}$
$P(\text{sample in bootstrap})=1-(1-\frac{1}{n})^n\to1-\frac{1}{e}\approx0.632$
$\Rightarrow\hat{R}_{(.632)}=0.368\cdot\hat{R}_{(A)}+0.632\cdot\hat{R}_{\mathrm{bs}}$
OLSE: $\hat{\beta}|X\sim\mathcal{N}(\beta^*,\sigma^2(X^\top X)^{-1})$ with $\hat{\sigma}^2=\frac{1}{n-d}\sum_i(X\hat{\beta}-Y)^2$.
$\Rightarrow1-\alpha$ CI: $\hat{\beta}_j\pm z_{\alpha/2}\widehat{\mathrm{se}}(\hat{\beta}_j)$
P-Value: $p=\sup_{\theta\in\Theta_0}\Pr[t(X_1,\dots X_n)\geq t(x_1,\dots,x_n)|\theta]$
Wald Test: Let $\hat{\theta}$ asymp. normal, then $W=\frac{\hat{\theta}-\theta_0}{\widehat{\mathrm{se}}}$

4.5 Bayesian Deep Learning

(Intr.) Posterior: $p(w|D)\propto p(w)p(D|w)$.
Var. Inf.: Find $q(w|\theta)=\mathcal{N}(\mu,\nu^2I)$
$\theta^*=\mathrm{argmin}_\theta\,\mathrm{KL}(q(w|\theta)\|p(w|D))=\mathrm{argmin}_\theta\,\mathbb{E}[w\sim q(\cdot|\theta)]F(w,\theta)$
where $F(w,\theta)=\log q(w|\theta)-\log p(w)-\log p(D|w)$.
Reparam.: $w=\mu+\nu\epsilon$, $\epsilon\sim\mathcal{N}(0,I)$
$\mu\leftarrow\mu-\eta_t(\epsilon^\top\frac{\partial}{\partial w}F(w,\phi)+\frac{\partial}{\partial\mu}F(w,\phi))$
$\nu\leftarrow\nu-\eta_t(\epsilon^\top\frac{\partial}{\partial w}F(w,\phi)+\frac{\partial}{\partial\nu}F(w,\phi))$

4.6 Active Learning

$I(X;Y) = H(X) - H(X|Y)$
Mutual Info: $I(X;Y) = \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)}$
Transd.-L.: $x_n = \text{argmax}_{x \in S} I(f_{x \in A}; y_x | D_{n-1})$
$\rightarrow$ For $f \sim \text{GP}(\mu, k)$: $I = \frac{1}{2} \log \frac{\text{Var}(y_x|D_{n-1})}{\text{Var}(y_x|f_{x\in A}, D_{n-1})}$
Safe BO: GP s_{f^*}, g^* observe $y_i = f^*(x_i)$, $z_i = g^*(x_i)$
95% conf.: $[l_n^f, u_n^f], [l_n^g, u_n^g]$
$S_n = \{x : l_n^g(x) \geq 0\}$, $\hat{S}_n = \{x : u_n^g(x) \geq 0\}$
$A_n = \{x \in \hat{S}_n : u_n^f(x) \geq \max_{x' \in S_n} l_n^f(x')\}$
Now ITL on sample space S_n, target space A_n.
Batch Active Learning: $B_\delta(x) = \{x' : \|x - x'\| \leq \delta\}$
$\Rightarrow \text{argmax}_{L \subseteq X, |L|=b} \Pr(\cup_{x \in L} B_\delta(x))$
$\Rightarrow \text{argmax}_{L \subseteq X, |L|=b} \frac{1}{|X|} |\{x' : \|x' - x\| \leq \delta, x \in L\}|$
NP-hard Problem, greedy algo (ProbCover):
$L = \emptyset$, $E = \{(x,x') : \|x - x'\| \leq \delta\}$ For $i = 1, \ldots b$:
Add $x_i = \text{argmax}_{x \in X} |\{x' : (x,x') \in E, x' \in X\}|$
Update $L \leftarrow L \cup \{x_i\}$, $E \leftarrow E \setminus (\{x_i\} \times (B_\delta(x_i) \cap X))$

5 Support Vector Machine (SVM)

Primal (soft margin): $\min_{w,w_0,\xi} \frac{1}{2}\|w\|^2 + C \sum_i \xi_i$
s.t. $y_i(w^\top x_i + w_0) \geq 1 - \xi_i$ and $\xi_i \geq 0$
$\hookrightarrow$ intractable if $\phi(x_i)$ instead of x_i
$\hookrightarrow$ $\xi_i = 0$ means x_i was not neglected
Dual: $\max_\alpha \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j x_i^\top x_j$
s.t. $0 \leq \alpha_i \leq C$; $\sum_i \alpha_i y_i = 0$
$\hookrightarrow$ solve α via quadratic optimisation
Optimal hyperplane: $w^* = \sum_i \alpha_i^* y_i x_i$
$\hookrightarrow$ $\alpha_i^* \neq 0$ only for support vectors
Optimal slack: $\xi_i^* = \max(0, 1 - y_i(w^{*\top} x_i + w_0^*))$

5.1 Structural SVMs

$\min_{w,\xi} \frac{1}{2}\|w\|^2 + C \sum_{i \leq n} \xi_i$
s.t. $\xi_i \geq 0$ and $\forall z \neq z_i$:
$w^\top \Psi(z_i, y_i) - w^\top \Psi(z, y_i) \geq \Delta(z, z_i) + -\xi_i$
Ψ : joint-feature map; Δ : loss / class dissimilarity func.; $w^\top \Psi(x,y)$: compatibility score btw. $x \& y$
Prediction: $\hat{z} = \text{argmax}_z\, w^\top \Psi(z, y)$

6 Ensemble Methods

$\hat{f}(x) \triangleq \frac{1}{B} \sum_{i \leq B} \hat{f}_i(x)$
$\text{bias}[\hat{f}(x)] = \frac{1}{B} \sum \text{bias}[\hat{f}_i(x)]$
$\mathbb{V}\hat{f} = \frac{1}{B^2} \sum \mathbb{V}[D]\hat{f}_i + \frac{1}{B^2} \sum \sum_{i,j} \text{Cov}(\hat{f}_i, \hat{f}_j) \approx \frac{\sigma^2}{B}$

6.1 Bagging (Bootstrap aggregation)

1. Draw B bootstrap sets $Z^{*1}, \ldots, Z^{*B}$
2. Train B base models $c_1, \ldots, c_B$
3. Aggregate: $\hat{c}_{\text{BAG}}(x) = \begin{cases} \frac{1}{B} \sum_{i \leq B} c_i(x) & \text{regr.} \\ \text{sign}(\sum_i c_i(x)) & \text{class.} \end{cases}$

Why it works: Small variance (weak learners), small covariance (almost indep. since $Z^{*i} \neq Z^{*j}$).
Compare, then Bag: Train c_b', c_b'', choose $\hat{c}_{\text{BAG}}'(x)$ if $\text{median}(\Delta_1, \ldots, \Delta_B) > 0$ with $\Delta_b := \hat{R}_n(c_b'') - \hat{R}_n(c_b')$
Random Forest: At each splitting step, u.a.r. choose m of p features & split only one (best) feature. $\rightarrow$ reduce correlation between trees.

6.2 Boosting

Sequentially train weak learners on all data, but $\rightarrow$ weight of misclass. samples ($\rightarrow$ bias).
AdaBoost: Forward stagewise additive modeling with exp. loss: $J(F) = \mathbb{E}\exp(-yF(x))$, min. at $F(x) = \frac{1}{2} \log \frac{P(y=1|x)}{P(y=-1|x)}$ (log-odds).
Max-margin, self-avg. and interpolating ($\rightarrow$ overfitting) classifiers.
[Init]: $w_i \leftarrow 1/n\ \forall i \leq n$
for $b = 1 \ldots B$:
  [Train]: Fit classifier $c_b(x)$ using weights w_i
  [Eval]: $\epsilon_b = \frac{\sum_{i=1}^n w_i \mathbb{I}\{c_b(x_i) \neq y_i\}}{\sum_{i=1}^n w_i}$ // error of c_b
  [Update]: $\alpha_b = \log(\frac{1-\epsilon_b}{\epsilon_b})$ // always > 0, final weight of c_b
  [Reweight]: $w_i \leftarrow w_i \exp(\alpha_b \mathbb{I}\{c_b(x_i) \neq y_i\})$ normalize!
Return $\hat{c}_{\text{ADA}}(x) = \text{sign}(\sum_{b=1}^B \alpha_b c_b(x))$

7 Deep Learning

Sigmoid: $\sigma(x) = \frac{1}{1+\exp(-x)} = \frac{e^x}{e^x+1}$, $\sigma'(x) = \sigma(x)\sigma(-x)$
Softmax: $y_i \propto \exp(\beta z_i)$

7.1 Variational Autoencoders (VAE)

Def:
$p_{\theta'}(z) \xrightarrow{\text{prior}} Z \xrightarrow[\text{likelihood}]{\text{dec}_\theta(z) = p_\theta(x|z)} X \xrightarrow[\text{approx. posterior}]{\text{enc}_\phi(x) = q_\phi(z|x)} Z$
sample/obs. X_i from latent representation Z
Train: $\text{argmax}_{\theta',\theta,\phi} \sum_i \log p_{\theta',\theta}(x_i)$
$(*) = \mathbb{E}[Z \sim q_\phi(\cdot|x_i)] \log \frac{p_{\theta',\theta}(x_i, Z)}{p_{\theta',\theta}(Z|x_i)} \frac{q_\phi(Z|x_i)}{q_\phi(Z|x_i)}$
$= \underbrace{\mathbb{E}\log p_\theta(x_i|Z) - D_{\text{KL}}(q_\phi(\cdot|x_i) \| p_{\theta'}(\cdot))}_{L(x_i,\theta,\phi) \equiv \text{ELBO} = \text{Infomax} - \text{Regularisation term}} + \underbrace{D_{\text{KL}}(q_\phi(\cdot|x_i) \| p_{\theta',\theta}(\cdot|x_i))}_{\geq 0} \geq L(x_i, \theta, \phi)$
Train: $\theta^*, \phi^* = \text{argmax}_{\theta,\phi} L(x_i, \theta, \phi)$

- **informative:** $\theta^* = \text{argmax}_\theta I(X;Z) = \text{argmax}_\theta \mathbb{E}[X,Z] \log p(X|Z) -$ const w.r.t. $\theta \approx \text{argmax}_\theta \sum_i \mathbb{E}[Z|X] \log p(x_i|Z)$
- **disentangled:** components in Z associated with distinct feature in X (see D_{KL} in ELBO).
- **robust:** noise in Z doesn't substantially affect X (and vice versa). $\rightarrow$ choice of approx. post.!

7.2 Transformers

Attention: Query: $Q = EW^Q$, Key: $K = EW^K$, Value: $V = EW^V$ from embedding E.
$A = \text{softmax}(\frac{QK^\top}{\sqrt{d}})V$, d: E-dim.

- **Self-Attention:** Q, K, V from same sequence
- **Cross-Att.:** Q from target, K, V from source.
- **Multi-headed:** Multiple weights, Stack A_i's.
- **Masked Self-Attention:** Future tokens masked ($-\infty$ in upper triangle of $P = QK^\top$)

Pos. Encoding: $p_{i,2j} = \sin(\alpha^{-2j} i)$, $p_{i,2j+1} = \cos(\alpha^{-2j} i)$, for $j \leq d/2$, $\alpha = 10^{4/d}$.

7.3 Stable Diffusion

Forward Process: For $t \leq T$, $\beta_t > 0$, let $\alpha_t = 1 - \beta_t$, $\bar{\alpha}_t = \prod_{s \leq t} \alpha_s$
$q(x_t|x_{t-1}) = \mathcal{N}(x_t|\sqrt{1-\beta_t} x_{t-1}, \beta_t I)$
$q(x_t|x_0) = \mathcal{N}(x_t|\sqrt{\bar{\alpha}_t} x_0, (1-\bar{\alpha}_t) I)$
Reverse Process: UNet predicts noise $\epsilon_\theta(x_t, t)$ added to form x_t from x_0. Then reconstruct x_{t-1}:
$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\sqrt{1-\alpha_t}}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right) + \sigma_t z$, $z \sim \mathcal{N}(0, I)$
Stable Diffusion: Combines text-image encoder (CLIP) with UNet in latent space.
CLIP: Train text/image emb. E_t, E_i to max. similarity: $\max_{i=j} \text{sim}(E_i^t, E_j^i)$, $\min_{i \neq j} \text{sim}(E_i^t, E_j^i)$
Cross-Attention in UNet:
$P_{bfmhw} = \sum_d Q_{bfdhw} K_{bfdm}$
$S = s(P/\sqrt{d_K})$
$A_{bfdhw} = \sum_m S_{bfmhw} V_{bfmd}$

8 Clustering

k-Means: $R_{km}(c, Y) = \sum_{x \in X} \|x - \mu_{c(x)}\|^2$

8.1 Mixture Models

Assume: $x \sim p(x|\pi_{1\ldots k}, \theta_{1\ldots k}) = \sum_{c \leq k} \pi_c p(x|\theta_c)$
Find: $\hat{\theta} = \text{argmax}_\theta\, p(X|\pi, \theta) = \prod_x p(x|\pi, \theta)$
Gaussian Mixtures: $\rightarrow p(x|\theta_c) = p(x|\mu, \Sigma)$
Introduce latent indicator variables for mode assignments $M_{xc} \in \{0, 1\}$. Then, the log-likelihood:
$L(X, M|\theta) = \sum_x \sum_{c \leq k} M_{xc} \log(\pi_c p(x|\theta_c))$

8.1.1 EM-Algorithm for Gaussian Mixtures

E-step: Calculate
$Q(\theta; \theta^{(t)}) = \mathbb{E}[M] L(X, M|\theta)|X, \theta^{(t)} = \ldots$
$= \sum_x \sum_{c \leq k} (\mathbb{E}[M] M_{xc}|X, \theta^{(t)} \cdot \log \pi_c p(x|\theta_c))$
where $\gamma_{xc} = \frac{p(x|c,\theta^{(t)}) p(c|\theta^{(t)})}{p(x|\theta^{(t)})}$, $\sum_{c \leq k} \gamma_{xc} = 1$
M-step: $\theta^{(t+1)} \in \text{argmax}_\theta\, Q(\theta; \theta^{(t)})$
s.t. $\sum_c \pi_c = 1$. Solve via Lagrangian, yields
$\pi_c = \frac{1}{|X|} \sum_x \gamma_{xc}$, $\mu_c = \frac{\sum_x \gamma_{xc} x}{\sum_x \gamma_{xc}}$, $\sigma_c^2 = \frac{\sum_x \gamma_{xc}(x - \mu_c)^2}{\sum_x \gamma_{xc}}$

8.2 Non-parametric Bayesian Methods

$\text{Dir}(x|\alpha) = \frac{1}{B(\alpha)} \prod_{k=1}^n x_k^{\alpha_k - 1}$, $B(\alpha) = \frac{\prod_{k=1}^n \Gamma(\alpha_k)}{\Gamma(\sum_{k=1}^n \alpha_k)}$
Rewrite Finite mixture models:
$p(x) = \sum_{k=1}^K \pi_k p(x|\theta_k) = \int p(x|\theta) G(\theta) d\theta$
where $G(\theta) = \sum_{k=1}^\infty \pi_k \delta_{\theta_k}(\theta) \leftarrow$ discrete distr.
Stick-breaking process:
Draw $\theta_k \sim H$ and $\beta_k \sim \text{Beta}(1, \alpha)$ for $k = 1, 2, \ldots$
$\pi_k = \beta_k(1 - \sum_{k=1}^{k-1} \pi_i) \Rightarrow \pi = \{\pi_k\}_{k=1}^\infty \sim \text{GEM}(\alpha)$
$\Rightarrow \sum_{k=1}^\infty \pi_k \delta_{\theta_k}(\theta) = G(\theta) \sim \text{DP}(\alpha, H)$
Sample $\theta^{(1)}, \theta^{(2)}, \ldots$ from G. Denote $\theta^{(i)} = \theta_{k_i}$.
$\Rightarrow \theta^{(i)}, \theta^{(j)}$ with $k_i = k_j$ belong to same "cluster"
Chinese Rest. Process:
$P(\text{cust}_{n+1} \text{ joins table } \tau | P) = \begin{cases} \frac{|\tau|}{\alpha+n} & \text{if } \tau \in P, \\ \frac{\alpha}{\alpha+n} & \text{new table} \end{cases}$
$P(\text{partition } P) = \frac{\alpha^{|P|}}{\alpha^{(n)}} \prod_{\tau \in P} (|\tau| - 1)!$
expec. #clusters: $\mathbb{E}[1] = \sum_{i \leq N} \frac{\alpha}{\alpha+i} \sim O(\alpha \log N)$
De Finetti: $(X_1, \ldots, X_n)$ are inftly exchangable RVs if $P(X_1, \ldots, X_n) = \int \left(\prod_{i=1}^n p(X_i|G)\right) dP(G)$

8.2.1 Finite GMM

1. Cluster centers: $\mu_k \sim \mathcal{N}(\mu_0, \sigma_0)$
2. Prob's of clusters: $\pi_{1\dots K} \sim \mathrm{Dir}(\alpha_{1\dots K})$
3. Cluster assignments: $z_i \sim \mathrm{Categorical}(\pi_{1\dots K})$
4. Coordinates of data: $x_i \sim \mathcal{N}(\mu_{z_i}, \sigma_{z_i})$

8.2.2 DP Mixture Model (DP-GMM)

1. Cluster centers: $\mu_k \sim \mathcal{N}(\mu_0, \sigma_0)$, $k = 1, 2, \dots$
2. Prob's of clusters: $\pi = (\pi_1, \pi_2, \dots) \sim \mathrm{GEM}(\alpha)$
3. Cluster assignments: $z_i \sim \mathrm{Categorical}(\pi)$
4. Coordinates of data: $x_i \sim \mathcal{N}(\mu_{z_i}, \sigma)$, $i = 1 \dots N$

Fitting a DP-MM: Collapsed Gibbs sampler
$p(z_i = k | z_{-i}, x, \alpha, \mu) \propto$ Prior $\times$ Likelihood
$\propto \begin{cases} \frac{|x_{-i,k}|}{\alpha+N-1} p(x_i | x_{-i,k}, \mu) & \text{for existing } k \\ \frac{\alpha}{\alpha+N-1} p(x_i|\mu) & \text{otw.} \end{cases}$
$x_{-i,c} \triangleq \{x_j | z_j = c, j \neq i\}$ data assigned to clust. c

9 PAC Learning

Want: Distribution indep. error guarantees!
Expec./Gener. error: $R(\hat{c}_n) = P_{X,Y}(\hat{c}_n(x) \neq c(x))$
Empirical error: $\hat{R}_n(\hat{c}_n) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{\hat{c}_n(x_i) \neq y_i\}$

PAC learnable: A can learn a concept class C from H if, given a suff. large sample, it outputs a hypothesis that generalizes well w/ high prob.
(1) $0 < \epsilon < 1/2$, $0 < \delta < 1/2$, (2) $P_{X,Y}$ on $X \times \{0, 1\}$:
If $n \geq \mathrm{poly}(1/\epsilon, 1/\delta, \dim(X))$,
Then $P_{X,Y}(R(\hat{c}_n) - \inf_{c \in C} R(c) \leq \epsilon) \geq 1 - \delta$.
If A runs in time polynomial in $1/\epsilon$ and $1/\delta$, we say that C is efficiently PAC learnable.

9.1 VC Inequality

$P(\dots \geq \epsilon) \leq \dots \leq \delta$
Select ERM: $\hat{c}_n^* = \mathrm{argmin}_{c \in C} \hat{R}_n(c)$. Under unif. conv.:
$P(R(\hat{c}_n^*) - \inf_{c \in C} R(c) > \epsilon) \leq P\left(\sup_{c \in C} |\hat{R}_n(c) - R(c)| > \frac{\epsilon}{2}\right)$

- $|C|$ Finite: $P(\sup|\cdots| > \epsilon) \leq 2|C| \exp(-2n\epsilon^2)$
- $|C|$ Unbounded: $P(\cdots) \leq 9n^{\mathrm{VCC}} \exp(-\frac{n\epsilon^2}{32})$

9.2 Rectangle Learning

$P((R(\hat{c}_n^*) > \epsilon) \leq |C| \cdot (1-\epsilon)^n \leq |C| \cdot \exp(-n\epsilon) < \delta$
Union bound: $P(\cup_i T_i) \leq \sum_i P(T_i)$

10 Appendix

Complete the square: If $p(x) \propto \exp(-\frac{1}{2} x^\top A x + x^\top b)$, then $p(x) = \mathcal{N}(x | A^{-1} b, A^{-1})$

Constrained optimisation:
primal: $\min_x f(x)$ s.t. $g_i(x) = 0$; $h_j(x) \leq 0$
Lagrangian: with each $\alpha_j \geq 0$
$L(x, \lambda, \alpha) = f(x) + \sum_i \lambda_i g_i(x) + \sum_j \alpha_j h_j(x)$
Solve: $\frac{\partial L}{\partial x} = 0$; $g_i(x) = 0$; $\alpha_j \geq 0$; $h_j(x) \leq 0$
If Slater's cond. holds, $\exists x : g_i(x) = 0, h_j(x) < 0$, then we can solve the dual instead:
$\max_{\lambda, \alpha} \{\min_x L(x, \lambda, \alpha)\}$ s.t. $\alpha_j \geq 0$
Solve: $\frac{\partial L}{\partial x} = 0$; $\frac{\partial L}{\partial \lambda} = 0$; $\alpha_j h_j(x) = 0$; $\alpha_j \geq 0$
Metrics:
acc $= \frac{TP+TN}{n}$, prec $= \frac{TP}{TP+FP}$, FPR $= \frac{FP}{FP+TN}$
Recall/TPR $= \frac{TP}{TP+FN}$, balanced acc $= \frac{1}{n} \sum_i \mathrm{TPR}_i$
F1-score $= \frac{2TP}{2TP+FP+FN}$, ROC = FPR/TPR
Conditional Gaussians:
$P_{X,Y} = \begin{bmatrix} X \\ Y \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu_X \\ \mu_Y \end{bmatrix}, \begin{bmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{YX} & \Sigma_{YY} \end{bmatrix}\right)$, Σ_{ij} p.s.d.
$\Rightarrow Y|X \sim \mathcal{N}(\tilde{\mu}, \tilde{\Sigma})$, where $\tilde{\mu} = \mu_Y + \Sigma_{YX} \Sigma_{XX}^{-1} (X - \mu_X)$,
$\tilde{\Sigma} = \Sigma_{YY} - \Sigma_{YX} \Sigma_{XX}^{-1} \Sigma_{XY}$
Maximum Mean Discrepancy: Is $p = q$?
$\mathrm{MMD}[\mathcal{F}, p, q] = \sup_{f \in \mathcal{F}} (\mathbb{E}[X \sim p] f(X) - \mathbb{E}[Y \sim q] f(Y))$
$= \|\mu_p - \mu_q\|$. $p = q \Leftrightarrow \mathrm{MMD}[\mathcal{F}, p, q] = 0$.
$\rightsquigarrow \mathrm{MMD}^2 = \mathbb{E}k(X, X') - 2\mathbb{E}k(X, Y) + \mathbb{E}k(Y, Y')$